Analysis I

Uebungsblatt 1

Problem:	Comment:	Points:
1		33.5 / 35

1

(a) $(A \cup B)^c = A^c \cap B^c$

Goal: Prove equality by double inclusion. Complements are taken relative to the universal set X.

($\subseteq$) Let $x \in (A \cup B)^c$. Then $x \notin A \cup B$, so $x \notin A$ and $x \notin B$. Hence $x \in A^c$ and $x \in B^c$, thus $x \in A^c \cap B^c$.

($\supseteq$) Let $x \in A^c \cap B^c$. Then $x \notin A$ and $x \notin B$, so $x \notin A \cup B$, i.e. $x \in (A \cup B)^c$.

Therefore $(A \cup B)^c = A^c \cap B^c$.

(b) $(A \cap B)^c = A^c \cup B^c$

($\subseteq$) Let $x \in (A \cap B)^c$. Then $x \notin A \cap B$, so $x \notin A$ or $x \notin B$. Thus $x \in A^c$ or $x \in B^c$, hence $x \in A^c \cup B^c$.

($\supseteq$) Let $x \in A^c \cup B^c$. Then $x \notin A$ or $x \notin B$, so $x \notin A \cap B$, i.e. $x \in (A \cap B)^c$.

Thus $(A \cap B)^c = A^c \cup B^c$.

(c) $A \setminus (A \setminus B) = A \cap B$

Let $x \in A \setminus (A \setminus B)$. Then $x \in A$ and $x \notin A \setminus B$. The condition $x \notin A \setminus B$ means it is not true that ($x \in A$ and $x \notin B$). Since $x \in A$ already, we must have $x \in B$. Hence $x \in A \cap B$; so $A \setminus (A \setminus B) \subseteq A \cap B$.

Conversely, if $x \in A \cap B$, then $x \in A$ and $x \in B$. Thus $x \notin A \setminus B$ (because $A \setminus B$ requires $x \notin B$). Therefore $x \in A \setminus (A \setminus B)$. Hence $A \cap B \subseteq A \setminus (A \setminus B)$.

So $A \setminus (A \setminus B) = A \cap B$.

2

(a)

Let $f : M \to N$ and $A, B \subseteq N$. We prove

$$f^{-1}(A \cap B) = f^{-1}(A) \cap f^{-1}(B), \qquad f^{-1}(A \cup B) = f^{-1}(A) \cup f^{-1}(B).$$

Intersection.

$$\begin{aligned} x \in f^{-1}(A \cap B) &\iff f(x) \in A \cap B \\ &\iff f(x) \in A \text{ and } f(x) \in B \\ &\iff x \in f^{-1}(A) \text{ and } x \in f^{-1}(B) \\ &\iff x \in f^{-1}(A) \cap f^{-1}(B). \end{aligned}$$

Thus the sets are equal.

Union.

$$\begin{aligned} x \in f^{-1}(A \cup B) &\iff f(x) \in A \cup B \\ &\iff f(x) \in A \text{ or } f(x) \in B \\ &\iff x \in f^{-1}(A) \text{ or } x \in f^{-1}(B) \\ &\iff x \in f^{-1}(A) \cup f^{-1}(B). \end{aligned}$$

Thus the sets are equal.

Conclusion. Preimages distribute over finite intersections and unions without any further condition on f.

(b)

Let $f : M \to N$ and $A, B \subseteq M$.

($\subseteq$): Let $y \in f(A \cap B)$. Then there exists $x \in A \cap B$ with $f(x) = y$. Hence $y \in f(A)$ and $y \in f(B)$, so $y \in f(A) \cap f(B)$. Therefore $f(A \cap B) \subseteq f(A) \cap f(B)$.

Equality does not always hold: For example: $M = \{1, 2\}$, $f(1) = f(2) = a$, $A = \{1\}$, $B = \{2\}$. Then $f(A \cap B) = \varnothing$, but $f(A) \cap f(B) = \{a\}$.

For the union: $f(A \cup B) = f(A) \cup f(B)$.

($\subseteq$) If $y \in f(A \cup B)$ then there is $x \in A \cup B$ with $f(x) = y$. So $x \in A$ or $x \in B$, hence $y \in f(A)$ or $y \in f(B)$. Thus $y \in f(A) \cup f(B)$.

($\supseteq$) If $y \in f(A) \cup f(B)$ then $y \in f(A)$ or $y \in f(B)$. In the first case $\exists x \in A$ with $f(x) = y$; but then $x \in A \subseteq A \cup B$, so $y \in f(A \cup B)$. The second case is analogous. So $f(A) \cup f(B) \subseteq f(A \cup B)$.

Combining both directions gives equality.

(c)

($\to$): Suppose f is injective. We always have $f(A \cap B) \subseteq f(A) \cap f(B)$. Let $y \in f(A) \cap f(B)$. Then there exist $x_1 \in A$, $x_2 \in B$ with $f(x_1) = f(x_2) = y$. Since f is injective, $x_1 = x_2 \in A \cap B$, so $y \in f(A \cap B)$. Thus $f(A) \cap f(B) \subseteq f(A \cap B)$.

($\Leftarrow$): Suppose f is not injective. Then there exist $x_1 \neq x_2$ with $f(x_1) = f(x_2)$. Let $A = \{x_1\}$, $B = \{x_2\}$. Then $f(A \cap B) = \varnothing$, but $f(A) \cap f(B) = \{f(x_1)\} \neq \varnothing$. So equality fails.

Therefore, f is injective if and only if for all $A, B \subseteq M$ we have

$$f(A \cap B) = f(A) \cap f(B).$$

3

(a)

Relation: For $A, B \subseteq \mathbb{N}$,

$$A \sim_1 B \iff A \setminus B \text{ is finite.}$$

Reflexive: For every A, $A \setminus A = \varnothing$, and $\varnothing$ is finite. Hence $A \sim_1 A$.

Not symmetric: Take $A = \{1\}$, $B = \mathbb{N}$. Then

$$A \setminus B = \varnothing \text{ (finite)} \quad \Rightarrow \quad A \sim_1 B,$$

but

$$B \setminus A = \mathbb{N} \setminus \{1\} \text{ is infinite} \quad \Rightarrow \quad B \not\sim_1 A.$$

So symmetry fails.

Transitive: Suppose $A \sim_1 B$ and $B \sim_1 C$; i.e. $A \setminus B$ and $B \setminus C$ are finite. If $x \in A \setminus C$, then either $x \in B$ (so $x \in B \setminus C$) or $x \notin B$ (so $x \in A \setminus B$). Thus

$$A \setminus C \subseteq (A \setminus B) \cup (B \setminus C).$$

A subset of a finite union of finite sets is finite. Therefore $A \setminus C$ is finite, i.e. $A \sim_1 C$.

Conclusion: $\sim_1$ is reflexive and transitive, but not symmetric; it is therefore a preorder (and not an equivalence relation).

(b)

Relation: For $A, B \subseteq \mathbb{N}$,

$$A \sim_2 B \iff \mathbb{N} \setminus (A \cap B) \text{ is finite.}$$

Not reflexive: Take $A = \{\text{even naturals}\}$. Then $\mathbb{N} \setminus (A \cap A) = \mathbb{N} \setminus A$ is infinite, so $A \not\sim_2 A$.

Symmetric: Since $A \cap B = B \cap A$, we have $\mathbb{N} \setminus (A \cap B)$ is finite if and only if $\mathbb{N} \setminus (B \cap A)$ is finite.

Transitive: Suppose $A \sim_2 B$ and $B \sim_2 C$; i.e. $A \cap B$ and $B \cap C$ are cofinite. Their intersection

$$(A \cap B) \cap (B \cap C) = B \cap A \cap C$$

is cofinite (intersection of cofinite sets is cofinite). Since $B \cap A \cap C \subseteq A \cap C$, the set $A \cap C$ contains a cofinite subset, so $\mathbb{N} \setminus (A \cap C)$ is finite. Thus $A \sim_2 C$.

Conclusion: $\sim_2$ is symmetric and transitive, but not reflexive.

$\mathbb{N} \setminus A \cap C \subset \mathbb{N} \setminus A \cup \mathbb{N} \setminus C \subset \mathbb{N} \setminus (A \cap B) \cup \mathbb{N} \setminus (B \cup C)$

(c)

Relation: For $A, B \subseteq \mathbb{N}$,

$$A \sim_3 B \iff \mathbb{N} \setminus (A \cup B) \text{ is finite.}$$

Not reflexive: Take $A = \{\text{even naturals}\}$. Then $\mathbb{N} \setminus (A \cup A) = \mathbb{N} \setminus A$ is infinite, so $A \not\sim_3 A$.

Symmetric: Since $A \cup B = B \cup A$, we have $\mathbb{N} \setminus (A \cup B)$ is finite if and only if $\mathbb{N} \setminus (B \cup A)$ is finite.

Not transitive: Take $A = \{1\}$, $B = \mathbb{N} \setminus \{1\}$, $C = \{2\}$. Then $A \cup B = \mathbb{N}$ and $B \cup C = B$ are cofinite, so $A \sim_3 B$ and $B \sim_3 C$. But $A \cup C = \{1, 2\}$ is not cofinite, hence $A \not\sim_3 C$. So transitivity fails.

Conclusion: $\sim_3$ is symmetric but neither reflexive nor transitive.

4

Let $M = \{1/n : n \in \mathbb{N} \wedge n \geq 1\} = \{1, 1/2, 1/3, 1/4, \ldots\}$.

Finding the greatest lower bound s and least upper bound t of M.

Claim: $s = 0$ and $t = 1$.

Proof that $t = 1$ is the least upper bound:

(i) 1 is an upper bound: For any $m \in M$, we have $m = 1/n$ for some $n \geq 1$. Since $n \geq 1$, we get $1/n \leq 1$, so $m \leq 1$.

(ii) 1 is the least upper bound: Suppose u is any upper bound of M. Since $1 \in M$ and u is an upper bound, we have $1 \leq u$. Therefore 1 is the smallest upper bound.

Proof that $s = 0$ is the greatest lower bound:

(i) 0 is a lower bound: For any $m \in M$, we have $m = 1/n$ for some $n \geq 1$. Since $n \geq 1 > 0$, we get $1/n > 0$, so $m > 0 \geq 0$.

(ii) 0 is the greatest lower bound: Let ℓ be any lower bound of M. We show $\ell \leq 0$.

Suppose for contradiction that $\ell > 0$. Since $1/n \to 0$ as $n \to \infty$, there exists $N \in \mathbb{N}$ such that $1/N < \ell$. But $1/N \in M$ and ℓ is a lower bound of M, so we must have $\ell \leq 1/N$. This contradicts $1/N < \ell$. (ARCHIMEDES PRINC)

Therefore $\ell \leq 0$, proving that 0 is the greatest lower bound.

Are $s, t \in M$?

For $t = 1$: Since $1 = 1/1$ and $1 \in \mathbb{N}$ with $1 \geq 1$, we have $t = 1 \in M$.

For $s = 0$: We need to check if $0 \in M$. For 0 to be in M, there would need to exist some $n \in \mathbb{N}$ with $n \geq 1$ such that $0 = 1/n$. But $1/n > 0$ for all $n \geq 1$, so there is no such n. Therefore $s = 0 \notin M$.

Answer: $t \in M$ but $s \notin M$.

5

(a)

Claim. For every $n \in \mathbb{N}$, the integer $n^7 - n$ is divisible by 7; equivalently $n^7 \equiv n \pmod 7$.

Proof (by induction on n).

Base case. for $n = 1$, $1^7 - 1 = 0$, divisible by 7.

Induction hypothesis. Assume for some $n \in \mathbb{N}$ that $7 \mid (n^7 - n)$; i.e. there exists an integer q with $n^7 - n = 7q$.

Induction step. We must show $7 \mid ((n+1)^7 - (n+1))$.

Expand $(n+1)^7$ using the binomial theorem:

$$(n+1)^7 = n^7 + 7n^6 + 21n^5 + 35n^4 + 35n^3 + 21n^2 + 7n + 1.$$

Subtract $(n+1)$:

$$\begin{aligned}(n+1)^7 - (n+1) &= n^7 + 7n^6 + 21n^5 + 35n^4 + 35n^3 + 21n^2 + 7n + 1 - n - 1 \\ &= n^7 + 7n^6 + 21n^5 + 35n^4 + 35n^3 + 21n^2 + 6n.\end{aligned}$$

Now subtract $(n^7 - n)$ (the part we know is a multiple of 7):

$$\big((n+1)^7 - (n+1)\big) - (n^7 - n) = 7n^6 + 21n^5 + 35n^4 + 35n^3 + 21n^2 + 7n = 7\big(n^6 + 3n^5 + 5n^4 + 5n^3 + 3n^2 + n\big),$$

which is clearly divisible by 7.

Conclusion. By induction, $7 \mid (n^7 - n)$ for all $n \in \mathbb{N}$.

(b)

Proof. Define $S_n = \sum_{k=1}^{n} (-1)^{k-1} k^2$. We prove

$$S_n = (-1)^{n-1} \frac{n(n+1)}{2} \qquad (n \geq 1)$$

by induction.

Base case $n = 1$. $S_1 = (-1)^0 1^2 = 1 = (-1)^0 \cdot \frac{1 \cdot 2}{2}$.

Induction step. Assume $S_n = (-1)^{n-1} \frac{n(n+1)}{2}$. Then

$$S_{n+1} = S_n + (-1)^n (n+1)^2 = (-1)^{n-1} \frac{n(n+1)}{2} + (-1)^n (n+1)^2 = (-1)^n \Big[-\frac{n(n+1)}{2} + (n+1)^2 \Big].$$

Factor $(n+1)$:

$$S_{n+1} = (-1)^n (n+1) \Big(-\frac{n}{2} + n + 1 \Big) = (-1)^n (n+1) \frac{n+2}{2} = (-1)^{(n+1)-1} \frac{(n+1)(n+2)}{2}.$$

Thus the formula holds for $n+1$, completing the induction. $\square$

(c) −1.5P $2^n \neq 2n$

Proof. Let $T_n = \sum_{k=1}^{2n} \frac{1}{k}$. We prove $T_n \leq n+1$ for all $n \geq 1$ by induction.

Base case $n = 1$. $T_1 = 1 + \frac{1}{2} = \frac{3}{2} \leq 2 = 1 + 1$.

Induction step. Assume $T_n \leq n+1$. Then

$$T_{n+1} = \sum_{k=1}^{2n+2} \frac{1}{k} = T_n + \frac{1}{2n+1} + \frac{1}{2n+2} \leq (n+1) + \frac{1}{2n+1} + \frac{1}{2n+2}.$$

Since $2n+1 \geq n+1$ and $2n+2 \geq n+1$ for $n \geq 1$,

$$\frac{1}{2n+1} + \frac{1}{2n+2} \leq \frac{1}{n+1} + \frac{1}{n+1} = \frac{2}{n+1} \leq 1.$$

Thus $T_{n+1} \leq (n+1) + 1 = n+2$. This completes the induction.

Therefore $\sum_{k=1}^{2n} \frac{1}{k} \leq n+1$ for all $n \geq 1$. □

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